Compiler Design - Unit Wise Topic Clustered Questions

UNIT 1: Compiler Basics & Lexical Analysis

Explain block diagram of phases of compiler.

Explain different phases of compiler in detail.

What are compiler writing tools? Explain in detail.

Explain bootstrapping and cross compiler with suitable example.

Define token. Find tokens in given program fragment.

Explain role of finite automata in lexical analysis.

Write LEX program for identifiers, keywords and operators.

UNIT 2: Syntax Analysis (Parsing)

Draw and explain LR parser. Differentiate LR(0), LR(1), LALR.

Construct LALR parsing table for given grammar.

Show grammar is LR(1) but not LALR.

Construct LL(1) parser and show moves.

Explain FIRST and FOLLOW sets.

Check whether grammar is ambiguous or not.

Explain handle and viable prefix.

UNIT 3: Semantic Analysis & Intermediate Code

Explain semantic rules and SDT with example.

Write SDT for evaluation of Boolean expression.

Give SDTS and three address code for program fragment.

Discuss different symbol table organization.

Explain inherited and synthesized attributes.

Write representations of intermediate code.

UNIT 4: Code Optimization

Explain peephole optimization technique.

Explain loop unrolling and loop jamming.

Explain DAG and its use in optimization.

Explain code motion and strength reduction.

List different loop optimization techniques.

UNIT 5: Code Generation & Runtime

Compute IN and OUT equations for given graph.

Explain storage allocation strategies.

Explain activation record and its need.

Explain issues in code generation phase.

Explain register allocation and target code generation.

Construct dominator tree and detect loops.